

Algebra II with Trigonometry

Chapter 11.2

Study guide

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[4666322_alg-2trig-h-chp-11-2-note-docx.pdf](#)

1. Short Summary

This document is a lesson from an Algebra 2 Trigonometry/Honors course that introduces ellipses. It discusses the geometric definition of an ellipse, its standard equations (horizontal and vertical), key components (foci, vertices, axes), the formula for eccentricity (a measure of how "stretched" the ellipse is), and several worked and practice problems. The lesson also includes examples of finding vertices, foci, axes, and writing equations given certain properties of ellipses.

2. Core Definitions, Theorems, and Formulas

Definitions

- **Ellipse:** The set of all points in a plane such that the sum of their distances from two fixed points (the foci) is a constant.
- **Foci (plural of focus):** The two fixed points used in the definition of the ellipse.
- **Vertices:** The points where the ellipse intersects its major axis.
- **Major Axis:** The longest diameter of the ellipse, passes through both foci.
- **Minor Axis:** The shortest diameter of the ellipse, perpendicular to the major axis.
- **Eccentricity (e):** A measure of how "stretched" an ellipse is.

Equations

- **Standard Form (center at origin)**

- **Horizontal major axis:**

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0$$

- **Vertical major axis:**

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, \quad a > b > 0$$

- **Endpoints**

- Major axis intersects at: $(\pm a, 0)$ or $(0, \pm a)$
- Minor axis intersects at: $(0, \pm b)$ or $(\pm b, 0)$

- **Foci Positions**

- For horizontal ellipse: $(\pm c, 0)$, where $c = \sqrt{a^2 - b^2}$
- For vertical ellipse: $(0, \pm c)$, where $c = \sqrt{a^2 - b^2}$

- **Lengths**

- Major axis length: $2a$
- Minor axis length: $2b$

- **Eccentricity**

- $e = \frac{c}{a}$ where $0 < e < 1$
- If $a = b$, the ellipse is a **circle** ($e = 0$)

3. Worked Study Guide

Step 1: Identify the type of ellipse

- Compare denominators. The larger value (a^2) tells you the axis that is major.
- Larger under x^2 : Horizontal ellipse.
- Larger under y^2 : Vertical ellipse.

Step 2: Find a, b, and c

- Equation in standard form will be either $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$, with $a > b$.

- $a = \sqrt{\text{larger denominator}}$
- $b = \sqrt{\text{smaller denominator}}$
- $c = \sqrt{a^2 - b^2}$

Step 3: Find Foci, Vertices, and Axes

- Vertices: $(\pm a, 0)$ or $(0, \pm a)$
- Foci: $(\pm c, 0)$ or $(0, \pm c)$
- Major axis: length $2a$
- Minor axis: length $2b$

Step 4: Calculate Eccentricity

- $e = \frac{c}{a}$
- The closer e is to 0, the more circular the ellipse; the closer to 1, the more elongated.

Step 5: Graph

- Plot the center, vertices, co-vertices (endpoints of minor axis), and foci.
- Sketch the smooth oval through these points.

4. Practice Problems with Answers

Problem 1

Given $\frac{x^2}{25} + \frac{y^2}{9} = 1$: 1. Find the vertices, foci, and axes lengths. 2. Find the eccentricity.

Answer: 1. $a^2 = 25 \implies a = 5$, $b^2 = 9 \implies b = 3$

- Vertices: $(\pm 5, 0)$
- Foci: $c = \sqrt{25 - 9} = 4$, so $(\pm 4, 0)$
- Major axis: $2a = 10$, Minor axis: $2b = 6$

2. $e = \frac{4}{5} = 0.8$

Problem 2

Write the standard equation of an ellipse with center at the origin, foci at $(0, \pm 3)$, and major axis length 8.

Answer:

- Major axis length $2a = 8 \implies a = 4$
 - $c = 3$, so $b^2 = a^2 - c^2 = 16 - 9 = 7$
 - Vertical ellipse: $\frac{x^2}{7} + \frac{y^2}{16} = 1$
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Problem 3

For the ellipse $\frac{x^2}{36} + \frac{y^2}{4} = 1$, what are the endpoints of the minor axis?

Answer:

- Minor axis: $2b = 4$, endpoints: $(0, \pm 2)$
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Problem 4

Find the eccentricity of the ellipse $\frac{x^2}{49} + \frac{y^2}{25} = 1$.

Answer:

- $a^2 = 49, b^2 = 25, a = 7, b = 5$
 - $c = \sqrt{49 - 25} = \sqrt{24} \approx 4.90$
 - $e = \frac{4.90}{7} \approx 0.70$
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Problem 5

Given ellipse $x^2 + 9y^2 = 36$, rewrite in standard form and find the major axis.

Answer:

- Divide by 36: $\frac{x^2}{36} + \frac{y^2}{4} = 1$
- Major axis under x^2 ($a^2 = 36$), $a = 6$
- Major axis length: $2a = 12$, so horizontal axis

5. Assumptions and Ambiguities

- Assumed all ellipses are centered at the origin unless otherwise noted.
- When "length of minor (or major) axis" is given, deduced a or b as half the given length.
- Some problems from the examples lack data for full solution (e.g., some from Example 2/3), so practice problems are newly written for clarity.
- Skipped graphing as this is text-based, but detailed the steps needed.
- All variables $a > b > 0$, and $c = \sqrt{a^2 - b^2}$ is only real and defined if $a > b$.

End of Guide